

What is an Ingress Controller?

- The Ingress Controller is responsible for load balancing and also controls the traffic that enters and exits the cluster.
- This controller is responsible for services that need to connect with other services that are located outside of the cluster.

ConfigMap

- To be more specific, it is a configuration file that may be edited by the user.

Secret

- It is similar to a configuration file, but designed to contain private data. It is Base64-encoded.

Volumes

- It is capable of retaining data forever and offering useful benefits.
- Because pods are ephemeral, it is necessary to store data in volumes so that pods of varied ages may continue to refer to the same state of the application.
- It provides capabilities that are analogous to that of an 'external hard drive', but for your pods.
- The volumes of your pods may either be stored locally on the same node or remotely on another node (e.g., cloud storage, NFS).

Deployment

- In most cases, we focus our attention on deployments rather than on pods.
- In case of problems with the main deployment, most also have a copy accessible as a backup.
- Stateful programs, on the other hand, can't have their components replicated, which means that things like databases can't be cloned either. We will need the Kubernetes module known as StatefulSet in order to do this. Due to the complexity of this undertaking, databases should almost always

be stored on servers that are geographically distinct from the Kubernetes cluster.

Utilisation of the Kubernetes platform

In order to continue with our introduction to Kubernetes, let's take up the *New York Times* as a case study.

- Following the publisher's decision to close its data centres, smaller deployments were carried out, and virtual machines (VMs) were placed in charge of managing the applications.
- They got to work in order to produce brand new instruments. However, at some time they realised that they were missing the target since they treated Amazon like any other data centre.
- The programmers quickly responded to the issue and came up with an elegant solution. The panel discussed the possibility of using Google Cloud Platform to deploy Kubernetes.
- The use of Kubernetes brought about a number of benefits, including the following:
 - A quickening of the pace in performance and delivery.
 - The time required for deployment was reduced from minutes down to seconds.
 - The updates were sent on their own and only when absolutely required.
 - Increased portability as well as an improved and more standardised approach of deployment for all engineers.

In conclusion, the *New York Times* has shifted from a system that was based on tickets to an automated system that is based on Kubernetes in order to request resources and schedule installs.

Kubernetes demo

In order to proceed with the tutorial on getting started with Kubernetes, let's

get directly into a demonstration.

1. Open the command prompt in Ubuntu.
2. The installation of all requirements may be completed with the following command:

```
$ sudo apt-get update
```

```
$ sudo apt-get install -y apt-transport-https
```

3. Make use of the following command in order to install the Docker dependency:

```
$ sudo apt install docker.io
```

4. To activate Docker and get it up and running, use the following commands:

```
$ sudo systemctl start
```

```
$ sudo apt-get install curl
```

Obtain the Kubernetes installation key, and then add it to your machine using the following instructions:

```
$ sudo curl -s https://packages.cloud.google.com/apt/doc/apt-key.gpg | sudo apt-key add
```

It is possible to change permissions by using the command that is listed here:

```
$ sudo chmod 777 /etc/apt/sources.list.d/
```

After that, create a repository by inserting the following information into the `/etc/apt/sources.list.d/kubernetes.list` file:

```
deb http://apt.kubernetes.io/
kubernetes-xenial main
```

Put your modifications in a safe place, and then close the file.

To configure Kubernetes, use the following commands: